Assessment 3: Group Project Report

Objective 2 – Linear Regression Models for FIFA World Cup 2026 Predicting Match Goal Difference and Individual Team Goals



Submitted by

Group 71

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Introduction:

In this report, two linear regression models are created for the 2026 FIFA World Cup. Linear Regression 2.1 is a model that predicts the goal difference between two opposing teams in a football match, based on 104 observations (1 row per match). Linear Regression 2.2 is used to predict the number of goals a team scores in a match, given a dataset of 208 observations (one row per match-team). The models use exactly 8 known pre-match variables, ensuring they are appropriate for real pre-match prediction.

Data Preparation:

Official datasets from FIFA (FIFA.org) were used to create two related datasets: one from FIFA Stats and the other from FBRef and The Stats Don't Lie. For Linear Regression 2.1, 104 rows were entered, one for each tournament match. The response variable is Goal_Difference. The eight pre-match predictors are Rank_Diff, Squad_Value_Diff, Rest_Days_Diff, Historical_Scoring_Diff, Elo_Rating_Diff, Top5_League_Player_Diff, Confed_Tier_Diff and Qualifying_GD_PerMatch. These factors reflect the relative strength, recent form and contextual differences between the two teams.

Linear Regression 2.2's data consisted of 208 team-match observations, with each match being represented by two rows. The dependent variable is the number of goals the team scores. The explanatory variables are Opponent_FIFA_Rank, Team_Squad_Value, Days_Rest, Team_Avg_Goals_Scored, Is_Host, Is_Knockout, Team_Avg_Goals_Conceded and Squad_Avg_Age. All variables are based on data prior to the match.All cleaning was done in Python. All empty rows were deleted; numeric columns were converted to suitable integer or float types, and the final cleaned file for team-match has no missing values. To ensure data quality and inform variable selection, exploratory graphs were created: goal distribution, goals versus FIFA rank, group stage versus knockout comparison, host status comparison, correlation heatmap, and actual versus pre-match average goals.

Model Specification and Estimation:

Both models were fitted using the least squares method in 'statsmodels'. An intercept term was added. Coefficient of determination (R^2), mean absolute error (MAE), and root mean squared error (RMSE) were used to assess model performance. To evaluate the classical linear model assumptions, residual diagnostics (residual-versus-fitted plots, Q-Q plots, Durbin-Watson statistic, and Breusch-Pagan test) were considered.

Goal Difference (Linear Regression 2.1). The model's R^2 was 0.41. The MAE was 1.08 goals, and the RMSE was 1.37 goals. Rank difference and historical scoring difference were the most significant, as it was expected that the teams' relative quality would play the biggest role in determining the outcome of the match.

Linear Regression – Goals Scored. The model had an R^2 of 0.37. The MAE and RMSE values were 0.94 goals and 1.22 goals, respectively. The most important factors for predicting accuracy were the team's average goals scored, the opponent's FIFA ranking, and knockout-stage status. The residual plots showed no noticeable systematic patterns, further supporting the linear specification for the available data.

Discussion:

The R^2 values represent a moderate level, which is typical of football prediction models. Match outcomes exhibit significant irreducible noise due to red cards, individual brilliance, referee decisions and other factors that are not directly measurable prior to kick-off. However, both models show that there is some meaningful information in the pre-match information. The team-match model (2.2) is especially relevant for understanding how a team's characteristics relate to goal scoring.

Some caveats include the use of approximate pre-tournament form indicators and the fact that only variables before the match can represent the dynamics of the match itself. Extensions might include more detailed ranking trajectories or new or recent windows of form of varying length.

Conclusion:

Two linear regression models satisfying the project requirements were constructed and evaluated. Both used information available before the match and included eight explanatory variables. This data can serve as a reliable basis for estimating goal differences and each team’s goals in the 2026 FIFA World Cup, demonstrating that relative ranking, scoring ability, and the context of recent matches are significant factors.

All Python scripts, cleaned datasets and exploratory graphs are available at:
https://github.com/DraigaPlatoon/DataScience_Assessment3